Expno: 3
Date

Understanding the Cisco Packet Tracer Environment

Aim:

To understand the Cisco Packet Tracer environment and design a simple computer network using its interface and networking components.

Steps:

1. Open the Cisco Packet Tracer application.
2. Observe the user interface, including the Menu Bar, Main Toolbar, Workspace, Common Tools Bar, Network component Box, and Realtime/Simulation tabs
3. From the Network component Box, select the required network devices such as PC, Hub, and Switch.
4. Drag and drop the devices into the workspace.
5. Connect the devices using Copper Straight-Through cables.
6. Configure IP addresses for the PCs using the Desktop -> IP configuration option.
7. Verify that all links are active
8. Use the Add Simple PDU tool to send a packet between two PCs
9. Observe packet transmission in Realtime mode and Simulation Mode.
10. Save the network topology

Result:

The Cisco Packet Tracer environment was successfully explored, and a simple network was designed. The interface components, device connections, IP configuration, and packet transmission were successfully verified

Experiment 1: HUB

Aim:

To study the behavior of a HUB by observing how it forwards data packets using Cisco Packet Tracer.

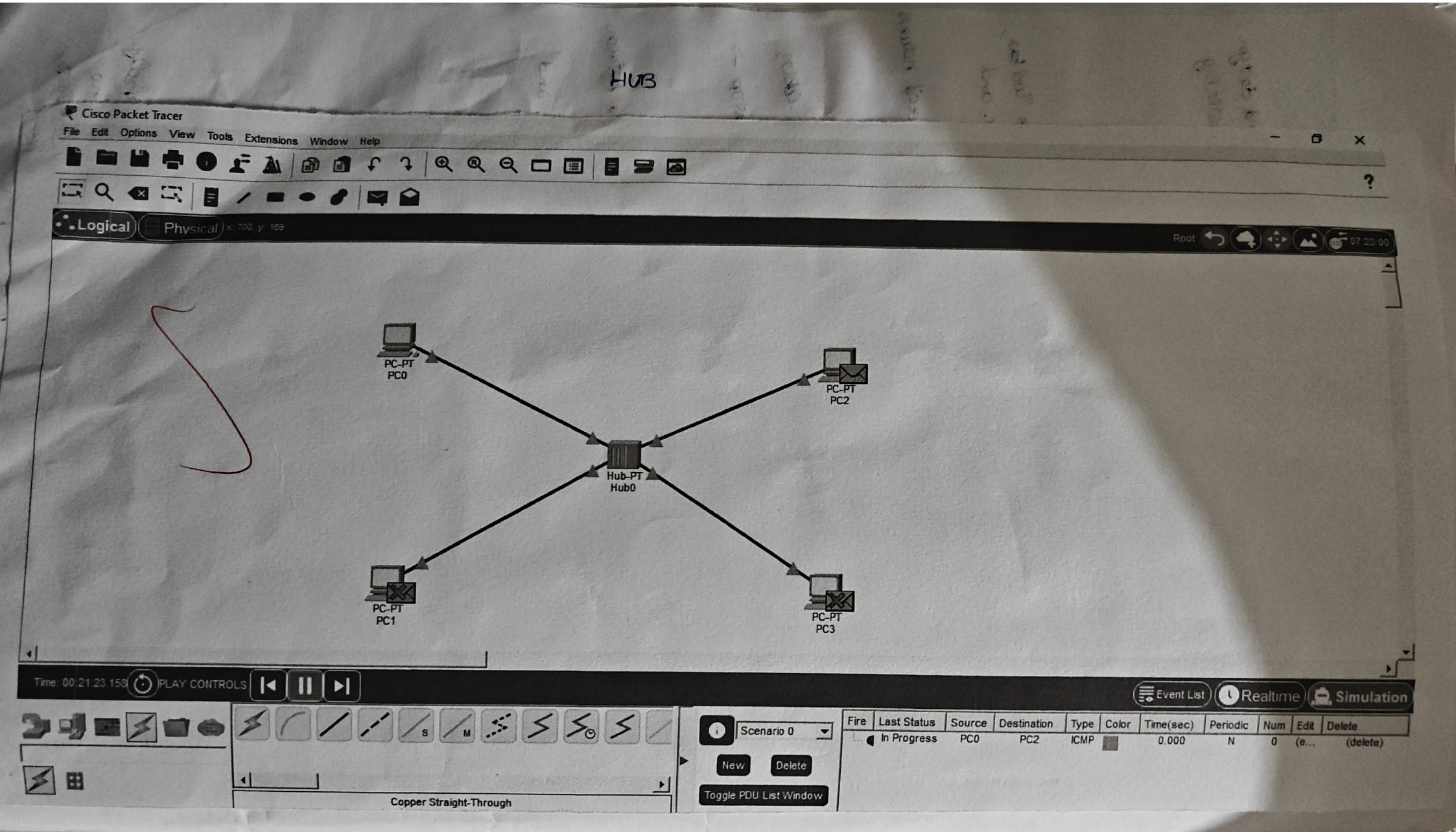
Steps:

1. Open Cisco packet Tracer
2. Drag and place 1 Hub and 4 Generic PCs into the workspace
3. Connect all PCs to the HUB using Copper Straight-Through cables
4. Configure IP addresses for all PCs in the same network.
5. Select Add Simple PDU from the Common Tools bar.
6. Send a packet from one PC (Source) to another PC (destination).
7. Switch to simulation Mode and observe how the Hub forwards the packet.

Result

The Hub broadcasts the received packet to all connected devices. Every PC receives the packet, but only the destination PC accepts it, while remaining others discard it.

HUB


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
PC-PT PC0
PC-PT PC2
Hub-PT Hub0
PC-PT PC1
PC-PT PC3
Time: 00:21:23.158
PLAY CONTROLS
Event List
Realtime
Simulation
Copper Straight-Through
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
In Progress
PC0
PC2
ICMP
0.000
N
0
(e...
(delete)

Experiment-2 : SWITCH

Aim :

To study the behavior of a switch by observing how it forwards data packets using Cisco Packet Tracer.

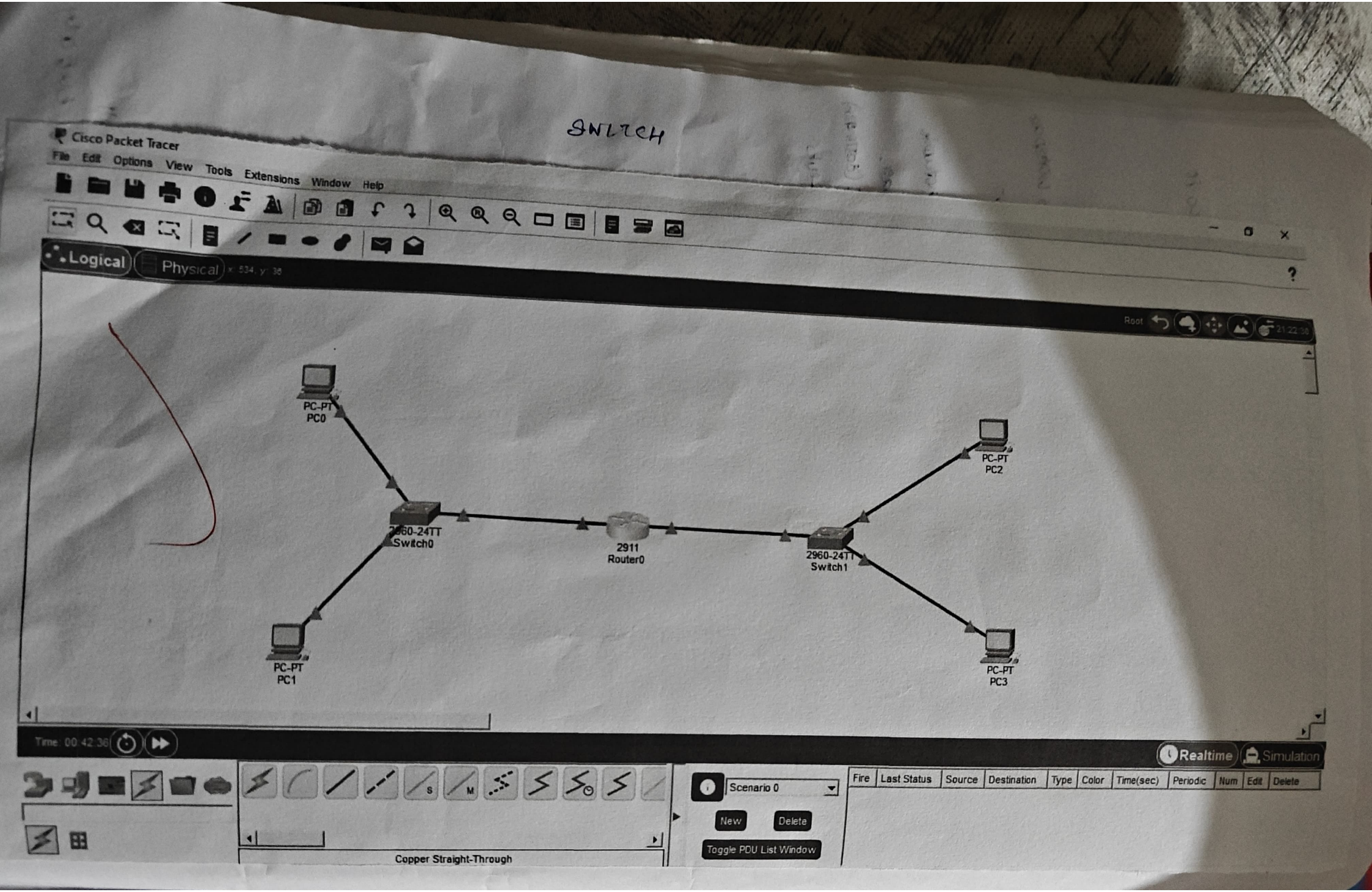
Steps

1. Open Cisco packet Tracer
2. Drag and place 1 switch and 4 Generic PCs into the workspace
3. Connect all PCs to the switch using Copper Straight-Through cables.
4. Configure IP addresses for all PCs in the same network
5. Select Add simple PDU from the Common Tools bar
6. Send a packet from one PC (source) to another PC (destination).
7. Switch to simulation mode and observe how the switch forwards the packet.

Result:

The switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency

SWITCH


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
PC-PT PC0
PC-PT PC1
2960-24TT Switch0
2911 Router0
2960-24TT Switch1
PC-PT PC2
PC-PT PC3
Time: 00:42:36
Realtime
Simulation
Copper Straight-Through
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete

Experiment 3: HUB and SWITCH

Aim

To compare the packet forwarding behavior of a Hub and a switch using Cisco Packet Tracer.

Steps:

1. Create one network using a Hub and another using a switch, each connected to 4 PCs.
2. Configure IP addresses for all PCs in both networks.
3. Send a simple PDU between two PCs in each network.
4. Observe packet transmission in Simulation mode.
5. Compare how the Hub and Switch forward packets.

Result:

The Hub broadcasts packets to all connected devices, whereas switch forward packets only to the intended destination. Therefore, the switch provides better performance, reduced network

HUB & SWITCH

